Lecture 08 — Angle and similarity

Mon Sep 14

Last updated: September 13, 2026.

Boyd video (optional): Lecture 12 — angle · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 3.4 (angle). Quiz 3 prep: Boyd 3.23, 3.24 (angle vs distance). Related objectives: Lectures. Schedule. Quiz 3 is Friday, September 18.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Check yourself**.

Learning objectives

By the end of class, students should be able to:

1. Compute cosine similarity or an angle between vectors.
2. Distinguish closeness in position from similarity in direction.
3. Choose an appropriate similarity measure for a stated purpose.
4. Explain why normalization may change a comparison.

Warm-up

Let $a = (1, 0)$ and $b = (1, 1)$.

(a) Compute $a^\mathsf{T} b$, $\|a\|$, and $\|b\|$.

(b) Without finding the angle in degrees, is $\cos\theta = a^\mathsf{T} b/(\|a\|\|b\|)$ closer to 1, to 0, or to -1? What does that suggest about the directions?

Fill in

For nonzero n-vectors a and b, the **angle** $\theta = \angle(a, b)$ satisfies

$$\cos\theta = \frac{a^\mathsf{T} b}{\|a\|\,\|b\|}, \qquad \theta \in [0, \pi].$$

The quantity $\dfrac{a^\mathsf{T} b}{\|a\|\,\|b\|}$ is the **cosine similarity**.

Cauchy–Schwarz says $|a^\mathsf{T} b| \le \|a\|\|b\|$, so the cosine is always between _____ and _____.

Interpretation: - $\theta \approx 0$ (cosine near 1): directions nearly the _____ - $\theta = \pi/2$ (cosine 0): vectors are _____ - $\theta \approx \pi$ (cosine near -1): directions nearly _____

Distance $\|a-b\|$ measures closeness of **positions**. **Angle** measures similarity of **directions** (independent of overall scale of each vector). Scaling a by a positive constant does _____ change $\angle(a,b)$.

A **normalized** (unit) vector is $\hat{a} = a/\|a\|$, which has $\|\hat{a}\| =$ _____. Among unit vectors, smaller distance is equivalent to _____ angle (Boyd 3.24(b)).

Software

Octave:

```
a = [1; 0]; b = [1; 1];
cos_th = (a'*b)/(norm(a)*norm(b))
th_deg = acosd(cos_th)    % angle in degrees
```

Python:

```
import numpy as np
a = np.array([1.0, 0.0]); b = np.array([1.0, 1.0])
cos_th = (a @ b) / (np.linalg.norm(a) * np.linalg.norm(b))
th_deg = np.degrees(np.arccos(np.clip(cos_th, -1, 1)))
```

Worked in class

W1. Compute the cosine and the angle (degrees is fine) between $u = (2,0)$ and $v = (1,1)$. Sketch both from the origin.

W2. Distance vs direction. Let $p = (1, 1)$, $q = (10, 10)$, and $r = (1, -1)$.

(a) Compute $\|p - q\|$ and $\|p - r\|$. Which is closer in **distance** to p?

(b) Compute $\angle(p, q)$ and $\angle(p, r)$. Which is more similar in **direction** to p?

(c) In one sentence: when would you prefer angle over Euclidean distance?

W3. Nonnegative vectors (Boyd 3.23). Let $x = (2, 1)$ and $y = (0, 3)$.

(a) Compute $x^\mathsf{T} y$. Why must the angle be between 0° and 90°?

(b) When are two nonzero nonnegative vectors orthogonal?

W4. Distance nearest neighbor vs angle nearest neighbor (Boyd 3.24(a)). Let

$$x = (1, 0), \qquad z_1 = (2, 0), \qquad z_2 = (0.5, 0.5).$$

Find the distance nearest neighbor of x among $\{z_1, z_2\}$ and the angle nearest neighbor. Are they the same?

W5. Normalization. Replace z_1 and z_2 by their unit versions $\hat{z}_1 = z_1/\|z_1\|$ and $\hat{z}_2 = z_2/\|z_2\|$. Does distance-NN of x among the unit vectors match angle-NN? (Boyd 3.24(b) idea.)

Check yourself

C1. Cosine of the angle between $(3, 4)$ and $(4, -3)$. Are they orthogonal?

C2. True or false: if $\|a - b\|$ is small, then $\angle(a, b)$ must be near 0. One counterexample or reason.

C3. Document word-count vectors (nonnegative). Why is angle a common similarity for documents?

C4. Octave or Python: write one line that prints the cosine similarity of `a` and `b` (assume they already exist).

C5. You normalize every candidate before using Euclidean distance to a query. What comparison are you effectively making?

Answers (Check yourself)

C1. $a^\top b = 12 - 12 = 0$, so cosine $= 0$. Yes, orthogonal (90°).

C2. **False.** Example: $a = (100, 0)$, $b = (101, 0)$ are close in distance but that alone does not force a general claim for arbitrary nearby pairs that point opposite ways after a shift; clearer counterexample for "small distance $\Rightarrow$ small angle": take $a = (0.01, 0)$ and $b = (-0.01, 0)$ — distance 0.02 is small, but angle is 180°.

C3. Documents can have very different lengths (total word counts). Angle / cosine ignores overall scale and focuses on the pattern of which words appear together.

C4. Octave: `(a'*b)/(norm(a)*norm(b))`. Python: `(a @ b)/(np.linalg.norm(a)*np.linalg.norm(b))`.

C5. Angle nearest-neighbor (among those candidates): after normalizing, Euclidean nearest matches smallest angle.